```
(from jedi>=0.18.2->ipython->ipython-sql) (0.8.7)
Requirement already satisfied: executing>=1.2.0 in .\anaconda3\Lib\site-packages (fr
om stack_data>=0.6.0->ipython->ipython-sql) (2.2.1)
Requirement already satisfied: asttokens>=2.1.0 in .\anaconda3\Lib\site-packages (fr
om stack_data>=0.6.0->ipython->ipython-sql) (3.0.1)
Requirement already satisfied: pure_eval in .\anaconda3\Lib\site-packages (from stac
k_data>=0.6.0->ipython->ipython-sql) (0.2.3)
Requirement already satisfied: numpy>=2.3.3 in .\anaconda3\Lib\site-packages (from p
andas->pyodc) (2.4.6)
Requirement already satisfied: python-dateutil>=2.8.2 in .\anaconda3\Lib\site-packag
es (from pandas->pyodc) (2.9.0.post0)
Requirement already satisfied: tzdata in .\anaconda3\Lib\site-packages (from pandas-
>pyodc) (2026.2)
Downloading ipython_sql-0.5.0-py3-none-any.whl (20 kB)
Downloading pyodc-1.5.0-py3-none-any.whl (31 kB)
Downloading findlibs-0.1.5-py3-none-any.whl (10 kB)
Downloading ipython_genutils-0.2.0-py2.py3-none-any.whl (26 kB)
Downloading prettytable-3.18.0-py3-none-any.whl (37 kB)
Downloading sqlparse-0.6.1-py3-none-any.whl (51 kB)
Installing collected packages: ipython-genutils, sqlparse, prettytable, findlibs, py
odc, ipython-sql

   ---------------------------------------- 0/6 [ipython-genutils]
   ------ --------------------------------- 1/6 [sqlparse]
   ------ --------------------------------- 1/6 [sqlparse]
   ------------ ------------------------- 2/6 [prettytable]
   ------------------------ ------------- 4/6 [pyodc]
   --------------------------------- ------ 5/6 [ipython-sql]
   ---------------------------------------- 6/6 [ipython-sql]

Successfully installed findlibs-0.1.5 ipython-genutils-0.2.0 ipython-sql-0.5.0 prett
ytable-3.18.0 pyodc-1.5.0 sqlparse-0.6.1
Note: you may need to restart the kernel to use updated packages.
```

In [1]:
```
%load_ext sql
```

In [4]:
```
%sql mssql+pyodbc://@DESKTOP-FO7MH95/Super_Store?driver=ODBC+Driver+18+for+SQL+Serv
```

In [8]:
```
%config SqlMagic.style = '_DEPRECATED_DEFAULT'
```

EDA

1- Sales by City

2- Transaction by Customers

3- Sales by Segment

4- Customer Segmentation based on Sales

Sales by City

In [14]:
```
%%sql
select top(10) City, round(sum(Sales),0) AS Total_Sales from Sample
group by City
order by sum(Sales) desc;
```

```
 * mssql+pyodbc://@DESKTOP-FO7MH95/Super_Store?TrustServerCertificate=yes&driver=ODB
C+Driver+18+for+SQL+Server&trusted_connection=yes
Done.
```

Out[14]:

City	Total_Sales
New York City	256368.0
Los Angeles	175851.0
Seattle	119541.0
San Francisco	112669.0
Philadelphia	109077.0
Houston	64505.0
Chicago	48540.0
San Diego	47521.0
Jacksonville	44713.0
Springfield	43054.0

Transaction by Customers

In [22]:
```
%%sql
select top(10) Customer_Name , count(Order_ID) as Number_of_Orders
from Sample
group by Customer_Name
order by count(Order_ID) desc;
```

```
 * mssql+pyodbc://@DESKTOP-FO7MH95/Super_Store?TrustServerCertificate=yes&driver=ODB
C+Driver+18+for+SQL+Server&trusted_connection=yes
Done.
```

Out[22]:

Customer_Name	Number_of_Orders
William Brown	37
John Lee	34
Matt Abelman	34
Paul Prost	34
Seth Vernon	32
Edward Hooks	32
Chloris Kastensmidt	32
Jonathan Doherty	32
Arthur Prichep	31
Zuschuss Carroll	31

Sales by Segment

In [52]:
```
%%sql
select Segment , round(sum(Sales),1) as Total_Sales
from Sample
group by Segment
order by sum(Sales) desc;
```

```
 * mssql+pyodbc://@DESKTOP-FO7MH95/Super_Store?TrustServerCertificate=yes&driver=ODBC+Driver+18+for+SQL+Server&trusted_connection=yes
Done.
```

Out[52]:

Segment	Total_Sales
Consumer	1161401.3
Corporate	706146.4
Home Office	429653.1

Customer Segmentation

In [40]:
```
%%sql
create view Customers_Levels as
select Customer_Name , round(sum(Sales),1) as Total_Sales,
case
 when sum(Sales) > 10000 then 'VIP_Customer'
 when sum(Sales) > 5000 then 'Average_Customer'
 else 'Normal Customer'
end as Customer_Levels
from Sample
group by Customer_Name;
```

```
 * mssql+pyodbc://@DESKTOP-FO7MH95/Super_Store?TrustServerCertificate=yes&driver=ODB
C+Driver+18+for+SQL+Server&trusted_connection=yes
Done.
```

Out[40]: []

In [51]:
```
%%sql
select Customer_Levels , count(Customer_Levels) as Customer_Count
from Customers_Levels
group by Customer_Levels
order by count(Customer_Levels) desc;
```

```
 * mssql+pyodbc://@DESKTOP-FO7MH95/Super_Store?TrustServerCertificate=yes&driver=ODB
C+Driver+18+for+SQL+Server&trusted_connection=yes
Done.
```

Out[51]:

Customer_Levels	Customer_Count
Normal Customer	676
Average_Customer	97
VIP_Customer	20